

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("data.csv")

print(df.head())
print(df.shape)
print(df.columns)
print(df.dtypes)

print(df.describe())

print(df.isnull().sum())

print(df.duplicated().sum())

corr = df.select_dtypes(include=np.number).corr()
plt.figure(figsize=(8,6))
plt.imshow(corr, cmap="coolwarm")
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()

df.hist(figsize=(12,8))
plt.suptitle("Histogram of Numerical Features")
plt.show()

numeric_cols = df.select_dtypes(include=np.number).columns
for col in numeric_cols:
    plt.figure(figsize=(6,4))
    plt.boxplot(df[col].dropna())
    plt.title(f"Boxplot of {col}")
    plt.show()

print("EDA Completed Successfully!")
print("- Checked missing values")
print("- Analyzed statistical summary")
print("- Found correlations")
print("- Visualized data distributions")
```

```
print("- Detected possible outliers")
```